🫁 Breathing, Exercise & Energy: Guided Notes

Biology A · Unit 5 · Level 3 (closer to the class work) · goes with Unit 3 Diabetes: A2 Exercise lab, A3 Combustion, A4 Yeast Feast, A5 Respiration + ATP, heart/lungs, A7

Name: _____

Use the class slides and your reading. Key words:

cellular respiration | *ATP* | *mitochondria* | *oxygen* | *carbon dioxide* | *combustion* | *yeast* | *BTB* | *heart rate*

A. Cellular respiration (A5)

1. Cellular respiration happens in the _____.

2. Fill in the table.

Inputs	Outputs
—	

3. ATP releases energy when a _____ group comes off.

B. Exercise lab (A2)

4. **Breathing rate change**

> Example: 14 breaths/min at rest, 28 after exercise → 28 − 14 = 14 more breaths per minute

a) Heart rate 70 at rest, 110 after exercise. Change = _____

b) BTB turned yellow in 90 s at rest and 45 s after exercise. How many times faster?

_____

5. What do the lab results show about exercise and CO2?

_____

_____

C. Combustion and yeast (A3, A4)

6. A burning marshmallow loses _____ because atoms leave as CO2 and water.

7. In the Yeast Feast, which tube made the most CO2 and why?

_____

_____